USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

1) Please fill out the Engineering Change Request Form on the following pages completely:
   a) Detail the names and contact details for each of the ECR contributors
   b) Update the ECR Title
   c) Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
2) For each section/table/figure to be updated:
   a) Detail the section number, starting page and figure/table number to be updated as appropriate.
   b) Detail existing text under "From Text"
   c) Detail changed text under "To Text"
3) Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step 1)b)
4) Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
   a) This ECR will then be reviewed by the Power Delivery Compliance Working Group.
   b) Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

USB PD CTS ENGINEERING CHANGE NOTICE

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: Structured VDM Lower Version Response Test
Applied to: USB PD CTS Q3, 2026

Brief description of the functional changes proposed:
Add a new test to verify SVDM Version check per base spec 8.5.1 Structured VDM Version rule.

Benefits as a result of the proposed changes:
Closes a normative-coverage gap in 8.5.1. This ECR adds field-level checks for backward compatibility with the devices reporting SVDM version lower that the UUT's highest supported version.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
Update per the ECR update.

An analysis of the Vendor Info File (VIF) implications:
None

USB PD CTS ENGINEERING CHANGE NOTICE

Actual Change Requested

TEST.PD.VDM.SNK.8 Structured VDM Lower Version Response

Description: The Tester verifies that on receipt of a Discover Identity with a Structured VDM Version below the UUT's highest supported version, a Sink-capable UUT replies using the same version it received.
Test Specific Tester Behavior: N/A
Test Conditions:

	Consumer Only, P/C, C/P, DRP	Provider Only
Rev2Snk	✓	
Rev3ChkdSnk	✓	

Test Procedures:

1. There are 2 possible bring-up procedures:
    a. The UUT has VIF field PD_Port_Type set to Provider/Consumer, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8
    b. The UUT has VIF field PD_Port_Type set to anything else, the Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2
2. The Tester sends a *Discover Identity* Command to the UUT, using SOP. The Structured VDM Version field (Major) and (Minor) in the VDM Header is set to 00b respectively.
3. The Tester checks the response [TEST.PD.VDM.SNK.8#1]
    a. If the UUT responds with either *Not_Supported* (in PD3 mode) or Ignore (in PD2 mode), the Tester checks that Supports_Structured_VDMs is NO. Test ends here.
    b. Otherwise, the check fails if a response message (ACK, NAK or BUSY) is not sent.
    c. The check fails if the Structured VDM Version (Major) and (Minor) field in the response VDM Header is not equal to the value sent by the tester in step 2 of this test.
4. In PD3 mode, Repeat steps 1-3 with Structured SVDM Version (Major) = 01b, (Minor) = 00b. [TEST.PD.VDM.SNK.8#2]

TEST.PD.VDM.SRC.3 Structured VDM Lower Version Response

Description: The Tester verifies that a Source-capable UUT replies to a Discover Identity using the same SVDM Version it received, when that Version is lower than its highest supported Version

Test Specific Tester Behavior: N/A

Test Conditions:

	Consumer Only	Provider Only, C/P, P/C, DRP
Rev2Src		✓
Rev3ChkdSrc		✓

Test Procedures:

1. There are 2 possible bring-up procedures:
    a. The UUT has VIF field PD_Port_Type set to Consumer/Provider, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.7.
    b. The UUT has VIF field PD_Port_Type set to anything else, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1.
2. The Tester sends *Request* Message with B25 (USB Communication Capable) set to 1b.
3. The Tester sends a *Discover Identity* Command to the UUT, using SOP. The Structured VDM Version field (Major) and (Minor) in the VDM Header is set to 00b respectively.
4. The Tester checks the response [TEST.PD.VDM.SRC.3#1]:
    a. If the UUT responds with either *Not_Supported* (in PD3 mode) or Ignore (in PD2 mode), the Tester checks that Supports_Structured_VDMs is NO. Test ends here.
    b. Otherwise, the check fails if the UUT does not return a Responder ACK / NAK / BUSY.
    c. The check fails if the Structured VDM Version (Major) and (Minor) field in the response VDM Header is not equal to the value sent by the tester in step 3 of this test.
5. In PD3 mode, Repeat steps 1-4 with Structured SVDM Version (Major) = 01b, (Minor) = 00b. [TEST.PD.VDM.SRC.3#2]